

Guidelines and conceptual model of costs for the public sector based on the Brazilian federal government's experience*

Nelson Machado**

Victor Branco de Holanda***

Summary: 1. Introduction and general aspects; 2. Schematic overview of the model and its main features; 3. Correlation between the concepts of government accounting and costs; 4. The relationship between public policies, funds, activities and cost objects; 5. Accumulation system and costing method in the public sector; 6. Costs in the public sector and government information systems; 7. Final remarks and conclusion.

Key words: costs in the public sector; conceptual framework of costs for the public sector; cost information systems; costing methods and cost accumulation system in the public sector.

* Article received in Feb. and accepted in May 2010.

** Doctor in accounting and controllership by the Faculdade de Economia, Administração e Contabilidade da Universidade de São Paulo (FEA/USP). Executive Secretary of the Ministry of Finance.

Address: Ministério da Fazenda, Esplanada dos Ministérios, Edifício Sede, 4 andar, Gabinete Brasília — CEP 70.048- 900, Brasília, DF, Brasil. E-mail: nelson.machado@fazenda.gov.br.

*** Doctor in accounting and controllership by the Faculdade de Economia, Administração e Contabilidade da Universidade de São Paulo (FEA/USP). Professor at the Federal University of Rio Grande do Norte (UFRN). Director of strategic management of the Ministry of finance.

Address: Ministério da Fazenda, Esplanada dos Ministérios, Edifício Sede, 4 andar, Gabinete Brasília — CEP 70048-900, Brasília, DF, Brasil. E-mail: victorbolanda@gmail.com.

RAP — rio de Janeiro 44(4): 791-820, Jul./ago. 2010

Guidelines and conceptual model of costs for the public sector based on the Brazilian federal government's experience

This article presents the Cost Information System (CIS) of the Brazilian federal government. It describes the taxonomic model and its main features, from its concepts to its purposes and user groups, and explores the correlation between the concepts of governmental accounting and cost accounting. It demonstrates the relationship between public policies, fund management, activities and cost objects. To this end, it uses the concepts of accumulation system, costing method and costs in the public sector, as well as government information systems. The article also presents the theoretic conceptual model and the guidelines of the design, and describes the implementation process, which is still ongoing. It finally looks into the future after the development stage of the CIS, and urges its readers to dive into the debate about measuring costs. The importance of CIS to improve the quality of spending in the public sector is highlighted.

1. Introduction and General Aspects

The introduction of a new approach for public management that prioritizes the measurement of federal government costs is one of the essential steps towards the transformation of paradigms that exist today about the role and importance of the public sector as a propelling agent of efficiency in the use of public and scarce resources. Public sector administrators need a set of management information to execute public policies with efficiency, efficacy and effectiveness. Thus, it is necessary to have a costs information system capable of assisting on typical decisions, such as buying or renting, manufacturing internally or outsourcing a given service or activity; or even to allow comparisons between costs of the same activity or service, when manufactured by different organizational units, aiming at assessing and improving performance. Moreover, cost information associated with the benefits of public policies should be the basis for the formulation of the budget proposal, the budget being a conductive that allows the execution of expenditures and the provision of the planned public services.

It is not the intention of this article to make extensive quotations, nor delving on methodology, since this was done in the theoretical works of Machado (2005) and Holanda

(2002), which – amongst other studies – served as a basis for the “experiment” here related. Prior to being a scientific enterprise, for the authors, this is meant to reinforce to the social sciences approach as well as the utilization of action research, drawn with the help of compared case studies techniques with a participative approach of many years of direct observations. The action research consists in a cycle of analysis, “found fact”, conception, planning, execution and another found fact or evaluation. And then, a repetition of this full cycle of activities, really a spiral of such cycles. The interventions are an experimental manipulation and the resolution of the problem is the goal; consequently, its scientific value must be measured in relation to the results and the transformation of reality achieved.

2. Schematic Oversight of the Model and its Main Characteristics

Despite legal determinations, the fact is that public administration until 2010 (be it the Executive, Legislative or Judiciary) did not use costs systems, except for isolated initiatives of some organs. The reason is that, up to that moment, there was no system or adequate methodology capable of being applied systematically and uniformly to all federal public administration.

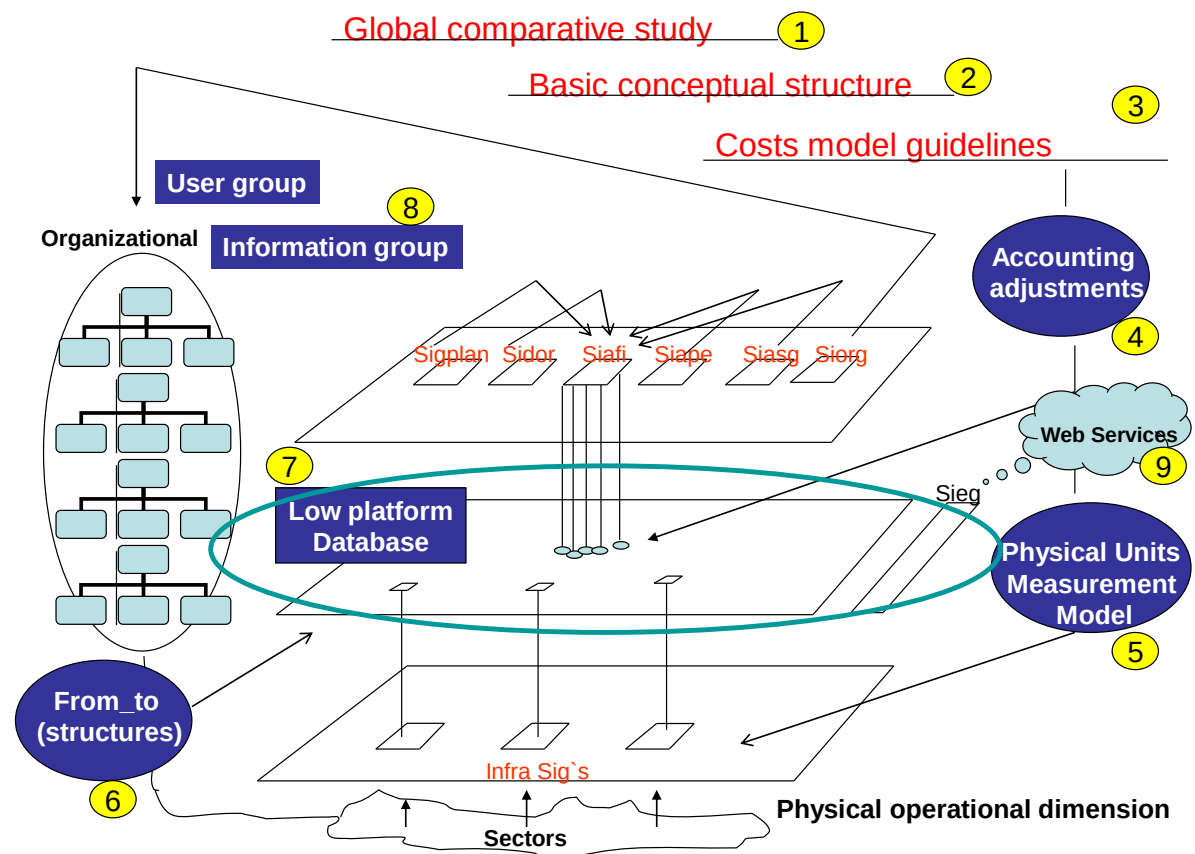
To fill this gap, in October 2008 the workshop “Cost Systems in Public Administration: conceptual model and implementation strategy” took place. In the event, the Brazilian Ministry of Finance was ratified as the organ responsible for coordinating the task of conceiving and implementing the Federal Government Cost Information System, by means of using data from the various structuring systems of the federal government.

As a consequence of the abovementioned decision, starting October 2008 the cost system had an executive project. It can be said that from that moment forward the topic stopped being, for the central organs of federal government, a matter of intention (will), leaving the sphere of discussion of ideas, to become a structured project with commissionaires, defined theoretical perspective, schedule and resources. These last events were determinant so that in March 2010 the federal government, through the National Treasury Office and the Serpro (Data Processing Federal Service), under the coordination of the Executive Office of

the Ministry of Finance, finally ratified the Cost Information System (CIS) of the Brazilian federal government.

To facilitate the exposition of some specific conceptual choices of the model adopted by the federal government, figure 1 is presented, which portrays the schematic oversight of the CIS model.

Figure 1
Oversight of the Model



Items “1” through “9”, plus the prototyping “10”, “11” implementation, “12” training, and “13” cultural change approach, form the set of components of the whole federal government CIS project. Part of the description of these components will be accomplished throughout this article, others, such as comparative study of international experiences and some considerations about the conceptual structure are objects of other works. Apart from presenting the model and its components, figure 1 will be useful for differentiating between

some important items of the model, for example the difference between the modeling by pilots *versus* the systemic and the concomitant.

Differences between Modeling by “pilots” versus the “systemic and concomitant”

In this topic, it is necessary first to emphasize that a cost information system conceived for the whole federal public administration (three powers) must guarantee the comparability attribute, but it must also allow the possibility of detailing information because of the different expectations and informational needs of the many organizational units and its respective users. The system must be capable of producing information in different levels of detail, by the generation of different reports, without losing the comparative attribute before mentioned. The most aggregated information must be generated by the system provided by the central organ, but the more analytical must be generated by the management information systems of each sectorial organ from the aggregated information, in other words, they must be reconcilable. Having made the separation between the dimension and span of a cost system for the whole federal administration, one can go into the specifics of the differentiation of the models proposed in this section.

After the many contributions and attempts at implementing of a cost system for the federal government, the main problem persisted: which model to adopt, how to build and implement it? In short, how to “make it happen”? There were important conceptual gaps and, mainly, divergences on the understanding of the flexibility and gradualism attributes when applied to the costs model. The gradualism and flexibility model previously proposed consisted first in the construction of many pilots in many distinct units of a cost system. This approach was modified and a new proposal was then drawn, which emphasized the discussion dealt in the theoretical works mentioned before.

An important point that ended up enabling the conception and then the implementation of CIS was the approach in two concomitant dimensions: one macro, systemic, generalizing and the other micro, specific, particular, that respects the physical operational specificity of each organ or entity. On the broad vision of the whole model, for its uniting characteristic, it was called a “bus station”, due to its basics fundamental characteristic of uniting the two

dimensions, the two “worlds”. On one hand, the deductive process of spending allocation to the objects of cost from the view of the central organs; on the other, the inductive process from the particular to the general, of the physical operational necessities of the sectorial finalistic organs (health, education, security, infrastructure etc.).

The first step of the model: the deductive – from the central organs via structuring systems to the “bus station” – has the advantage of generating synthetic cost information, however completely comparable for the whole federal public administration. This requisite makes CIS unique and very useful to evaluate the performance and result of public policies by the central organs of federal government. In the second step of the model (inductive: from the particular to the general), the system allows adjustments to each operational physical reality. In this phase, each administrative unit may perfect specific information received already adjusted to costs, provided in a uniform way, consistent and verifiable, referenced to common costs objects and standardized for the whole public administration. As mentioned, this fundamental requisite makes it viable the analysis of performance and results of public policies by the central organs of the federal government.

We highlight that this is the concept of gradualism which was adopted, i.e, the conception of a general model (1st level: deductive, until the “bus station”) concomitant to the specific approach (2nd level: inductive, from the “bus station” downwards) and not the approach of first the construction of pilots and then the general model. In sum, instead of “first” pilots and “then” the general, it was opted the “concomitant, all together”, according to the theoretical and systemic approach from the aforementioned theoretical works.

Definitions, purposes and users of the Cost Information System in the public sector

The Cost Information System of the Brazilian federal government (CIS) is a management support system, to be used both by the executor organs of public policies and by the planning and controlling central organs. The cost system is not another financial administration operational system or one that will substitute the current accounting systems. The cost systems do not have a transactional character, but a management one, establishing relations directed to the objects of cost, from information from other auxiliary systems.

Main purposes of the Cost Information System of the Brazilian federal government

The main purposes of the Cost Information System of the Brazilian federal government are:

- Increased efficiency of the allocation process – allows evaluating if certain goods or services produce by the government justify the corresponding cost, or if there are alternatives which have reduced costs, contributing to the evaluation of alternatives by the planning organs, usually during the decision-making process, as to the allocation of public resources (*ex ante* character);
- Increased operational efficiency – the production of goods and services by the state must submit itself to some evaluations, regarding the different ways of manufacturing the same product with lower cost, that allows the identification of operational inefficiencies and cost reduction opportunities; regarding the performance evaluation of people and organizations (including other entities and NGOs); regarding the identification of economies with current expenditures (CF, art. 39, §7th) and also regarding the cost/benefits of actions. The comparative analysis of the many different ways of making the same product with lower costs. Thus, it facilitates the identification of inefficiencies and cost reduction opportunities; results and performance analysis, as well as benchmarking and identification of low aggregated value activities;
- Fixation of public prices and rates – for calculating the sale price in governmental organizations that provide goods or services, knowing the cost of products is essential for establishing a fair price or for knowing the financial result, profit or loss.

Other purposes may also be highlighted, such as subsidizing the planning and budgeting process; supporting decisions on the continuity of the production of a certain good, service or activity, or handing it over to external entities; enabling the improvement of internal management information to enable the measurement of results and performances and provide control that is more oriented towards results. Knowing all the possibilities, the biggest challenge is making the generated information useful and appropriate to the decision making process, essentially to facilitate the task of the manager in search of maximizing the results of public resources of the society one is a part of.

Groups of Users of the Cost Information System of the Brazilian federal government

The success of the implementation of the Cost Information System of the Brazilian federal government by public entities does not depend only on legal demands nor the sophistication of its conceptual structure, it is fundamental that users see the utility of the cost information for the management control and improvement of one's activities and perfecting of the decision making process. Following, the main users of the Cost Information System of the Brazilian federal government are listed:

- National Treasury Office (STN) – central organ of the federal accounting system, responsible for the development and implementation of the Cost Information System of the Brazilian federal government;
- Federal Budget Office (SOF/MF) – coordinates the process of allocating public resources under the scope of the planning and budget federal system, and administers the Sidor;
- Executive Offices (Secex) – responsible for assisting the ministers of State in the supervision and strategic coordination of activities of the organs part of the structure of each ministry and entities bound to them.
- Planning, budget and Administration sub-offices (Spoas) – sectorial organs responsible for the operationalization of the systems for planning and federal budget, federal financial administration, federal accounting, federal civilian personnel administration, information and computers resources administration, general services and national archives, under the scope of each organ/entity;
- Strategic Investments and Planning Office (SPI/MF) – coordinates the planning sector of the Federal Planning and Budget System and manages the “Pluriannual Plan” and the SIGPlan;
- Management System for the Pluriannual Plan – unites a set of actors responsible in different levels for the application of public resources and the management of programs, actions and projects under the scope of each federal public organ/entity. In the current model, there is a program manager, and executive manager, an action coordinator, an executive coordinator, and an evaluation and monitoring unit;

- Expenditure officers¹ - responsible for the authorization of expenses under the scope of federal public organs/entities;
- Brazilian Inspectorate-general (CGU)² – responsible for evaluating governmental action, the management of federal public managers and the appliance of public resources by entities of private law, through accounting, financial, budgetary, operational and property inspection;
- Accountability Office (TCU) – helps the national congress on the external control, through the exercise of accounting, financial, budgetary, operational and property inspection of the Union and entities of the direct and indirect administration, as to legality, legitimacy and economy and inspection of the implementation of grants and the tax breaks;
- Society – researchers, specialized analysts and the citizen more broadly, responsible for the social control of public expenditure.

To understand the perception of utility and the need of cost information from the federal government user's perspective, as well as other important questions for the modeling of the system, a research was drawn in all of the federal public administration. The research was manipulated by questionnaires and the summed up analysis of the data was presented in the I International Seminar on Cost Information (Esaf/DF) and is available in the website: <www.socialiris.org/custosnosetorpublico>.

Basic Assumptions and Information about Cost Information

The generation of cost information assumes the adoption of accrual basis. As long as the governmental accounting does not adopt in its plenitude the accrual basis “accounting adjustments” will be necessary, according to component 4 of the general model (figure 1). As for the qualitative characteristics of cost information, those may be conceptualized briefly as

¹ Allocated, in occasion, in positions previously mentioned.

² Internal Control System for the Federal Executive power, integrated by: I - Inspectorate-general, as a central organ, responsible for the normative orientation and technical oversight of the organs that compose the system; II – Internal control offices (Cisets) of the Chief of Staff, the office of the solicitor general, of the ministry of foreign affairs and the ministry of Defense, as sectorial organs; and III – the internal control units of the military commands, as sectorial units of the Ministry of Defense Internal Control Office.

essential attributes for the information to be adequate. Those may be (without the pretension of exhaustion) listed as follows: understandability, timeliness, comparability, granularity, reliability.

3. Correlation of the concepts of governmental accounting and of costs

For the use of governmental accounting information to be possible in the construction of a costs system, a terminology and conceptual approximation between governmental and cost accounting is necessary. Analyzing the terminology of costs accounting, the following concepts are highlighted (Martins, 2003:25):

- Expenditure – “financial sacrifice with which the entity bears to obtain a certain good or service, this sacrifice being represented by the delivery or promise of delivery of assets”;
- Investments – “expense activated due to its service life or benefits attributable to future periods”;
- Cost – “expense related to a good or service used in the production of other goods and services”;
- Expense – “good or service consumed directly or indirectly for obtaining revenue”;
- Disbursement – “payment resultant of the acquisition of a good or service”.

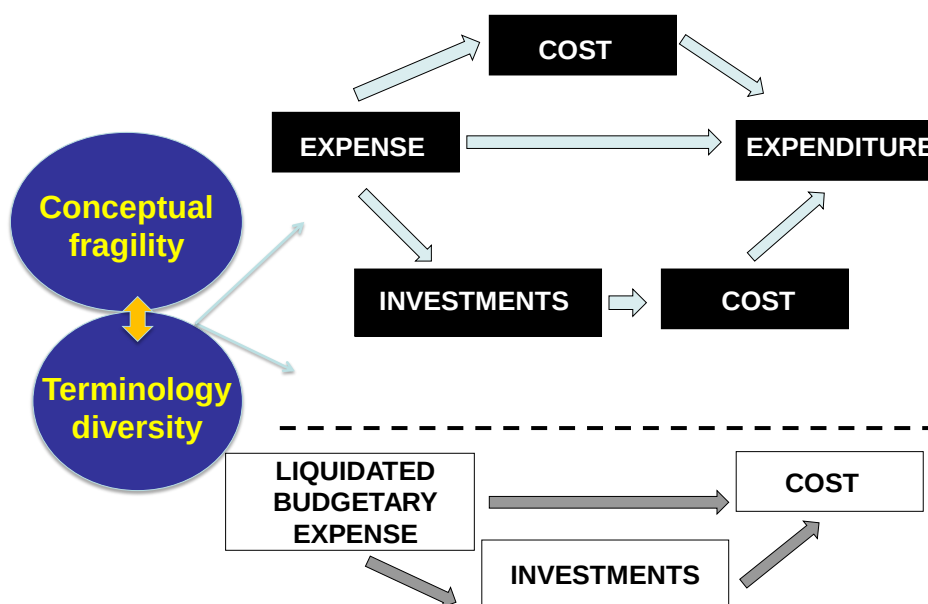
These concepts may be thought as a cycle: the (active) equipment used in the production process, which was spent and transformed itself in investment and was later considered partially as cost becomes an expense of the period in the process of determination of results. Of course, the expenditure can also transform itself immediately into expense or cost, without going through the investment phase. The most frequent is that expenditure distributes itself as investments, costs and expenses, except in a specific case, where characterized as “loss” for corresponding to reductions in patrimony that do not enable any compensatory receipt nor generated goods or services.

Figure number 2 shows this relationship and highlights the fact that still nowadays some difficulty on the application of these terms are observed in the process of analyzing expenditure in the public sector.

Figure 2

Oversight of the correlation of concepts in governmental and cost accounting

Processing Cycle: spent, investment, cost and expense



On business accounting, it is important to highlight the difference between cost and expense. Aiming at determining cost for the effect of stock evaluation, the accounting of traditional costs has been putting a relatively simple frontier between these two concepts, in other words, all expenses incurred in the product, until it is ready for sale, are production costs. From that point forward, expenses are considered expenditures. Therefore, the expenses with commercialization are considered expenditures and will impact on the results of the period. On the other hand, in the public sector, budgetary expenditures have a cycle of their own, going through the phases of fixation, effort, liquidation and payment.

Parallel between the business and the public sector

Tracing a parallel between the business and the public world, one can notice some similarities or correspondence between concepts. The fixation phase, in general terms, may be understood and practiced the same way as the establishment of the standard cost, since the standard cost is nothing but a kind of budget. The effort phase does not encounter correspondence in concepts usually used by costs accounting.

In the phase of liquidation, however, the criteria for expense execution are essentially verified and the “liquidated” values mean that the right to receive of the creditor was analyzed, as well as attested that the creditor fulfilled his contractual obligations. The payment phase, obviously, is identical to the concept of disbursement, observing only that, in public accounting, the payment must be preceded by all the previous phases, while in business accounting the disbursement might be done before, during or after the down payment of the utility purchased, so, lagged or not of the expenditure.

Table 1 shows the summary of this terminology correspondence.

Table 1

Governmental Accounting x Cost Accounting: terminology correspondence

governmental accounting	business costs accounting
fixation	standard
effort	not traditionally used
liquidation	expenditure (made)
payment	disbursement

Assuming, then, that the concepts “expenditure” and “liquidated” values are equivalent, one can consider the liquid budgetary expense as a starting point for the construction of the Cost Information System. The liquidated budgetary expenses will be costs when related to goods or services used and consumed in the current fiscal year, and investments when they

benefit the future exercise. The investments as in the private sector later transform themselves into costs. Figure 2 in the part below the dotted line shows this relationship.

One can then assert that in the public sector, given the current accounting model of government spending, the expenditure liquidated is the variable from which one should make the necessary conceptual adjustments to get to the cost information. Thus, in the public sector, the concept of liquidated expenditure works as a conceptual and methodological link to find out the costs.

It is important to highlight that, in the public sector, in contrast with the private sector, there is no need to distinguish between the cost of products and services and the expense of the period. This separation is very important when the objective of finding out the costs is the stock evaluation of the products that are going to be sold in the following period. In the public sector, this distinction is irrelevant (relativized), since the objective is finding out the cost of the public services rendered in the period and the assumption is the inexistence of a stock of goods and services or products which are processing and/or finished, fact for the large majority of the public sector.

The identification of investments from liquidated expenses is relatively simple. The classification of budgetary expenses on economical categories³ establishes immediately the distinction between current expenses and capital expenses, the investments being seconded in capital expenses.

The identification of costs from liquidated expenses may be done using the expenses classified as costing. Assuming that the objective of public entities is manufacturing goods and services in order to deliver them to the community and that the cost analysis is done within annual periods, the first approximation with governmental accounting leads to the assertion that all expenditures or liquidated values, classified as costing expenses, are costs of any good or service manufactured by the governmental entity in that period. However, for this assertion to be true some conceptual accounting adjustments are necessary.

³ The economics classification of expense has the following categories: current expenses – expense of costing and current transference. Capital expenses – investments, financial investments, wire transfer of capital.

Conceptual Accounting Adjustments

The costs model hereby proposed brings intrinsically the need of promoting an integration of budgetary, accounting, and cost information systems. That integration being possible, in today's reality, only by means of the accounting system and the classifications of budgetary expense. In the case of public services rendered by the state, there is no doubt that the liquidated expense is the one that is closest to the definition of expenditure as the expense with the acquisition of goods and services necessary to the production of other goods and services.

In liquidated values, there are quotas that need special treatment to become adequate to the concept of cost of goods and services of the period, that is the definition of adjustments from governmental accounting to the cost accounting here addressed. Some examples of these adjustments are: forced liquidation, purchase of materials for stocks and expenses of previous fiscal years.

The expenses with conservation work in real state may benefit more than one period and, therefore, are investments, but, since they are classified as costing expenses, it is necessary to reclassify them so as to not distort the cost of services rendered in the period.

The expenses with personnel of the entity, even when applied in investments, will always be classified in the group of current expenses. Therefore, it is necessary to identify them and to reclassify them.

The forced liquidation is registered in the closure of the fiscal year, having as counterpart the "unprocessed to be paid for excess" bill. Its objective is taking all expenses legally committed to compose the expenses of the financial exercise (art. 35, law n° 4.320). There are two basic causes that generate forced liquidation:⁴ the first is then services were rendered. They are, therefore, costs of the period, but there is no documentation able to process the normal liquidation. Examples of this situation are the water, energy and telephone services supplies for the month of December. In that case, since the forced liquidation is done by average, it is necessary to observe the value actually paid in the next fiscal year. If the value is lower than the liquidation, the canceling of one of the quotas of the unprocessed to be

⁴ The forced liquidation of works in progress is not addressed here because it does not interfere in the counting of services costs of the period. Besides, it also has no impact in activation of works when concluded, since that is done after the last measurement and by the value accumulated throughout the whole contract.

paid for excess will occur, and that quota should be used to the cost adjustments of analysis provision. If the value is higher than the amount of the forced liquidation, it is necessary to make a new effort using the allocation relative to the expenses of the previous fiscal years. The second refers to the supply of consumer goods in transit; since the merchandise has not arrived, it obviously cannot be consumed. Thus, it is a cost of the period. In this case, the most adequate treatment would be considering all purchases in transit as stock material, not impacting, then, the costs of the period as it will be seen in the next item.

In buying consumer goods, there must be a differentiation between those which are for immediate consumption and those that transit in warehouses, constituting, therefore, stock for some time. The purchases for immediate consumption don't offer great difficulty: once liquidated, they are expenditures that transform themselves immediately in costs. On the other hand, as consumer goods in stock (warehouse) are part of the fixed assets, the governmental accounting gives it special treatment. When liquidation is taking place a projection is usually done in the financial system – *budgetary expense to budgetary expense* to be paid. However, this event will also impact the patrimonial system, in which the liquidation is projected – *warehouse to active equity changes* (the acquisition of goods for the warehouse) and, in the moment of consumption, the projection is *passive equity changes* (liquidation of goods by consumption) *to warehouse*. Thus, the balance of the warehouse account will show, in the balance sheet, the existence of consumer goods, whilst the “liquidation of goods by consumption” account, from the “asset variation” group, will show materials used in the period. Thus, the adjustment, in this case, is simple: just substitute the amount of supply purchases for the group stock of budgetary expenses, for the value of materials effectively consumed registered in the “liquidation of goods by consumption” account.

Consolidation and Schematic Overview of the Accounting Conceptual Adjustments

Thus, the emergence of cost information starts from making adjustments in the information obtained on Siafi. The accounting adjustments were done in two steps. The initial adjustment constituted in the identification and the loading of information based on ledger accounts which bring accounting and non-accounting information, adjusting by adding the

asset values and subtracting the budgetary values which are not costs of the period, in accordance with the concepts of cost accounting.

The second step of the adjustment was the exclusion of information that were in the accounts abovementioned, but that did not compose cost information. This adjustment used management parameters disposed in checking accounts of ledger accounts inserted on the first step. From the conceptual modeling, coming from the theoretical reflection, it arrives at the practical repercussion in the CIS. In an attempt of synthesis, the modeling of conceptual accounting adjustments is presented on frame 2, already with effects on information systems and its modeling for application.

Table 2

Conceptual Adjustments Model

Executed budgetary expense	
	Liquidated expense
	Executed expense by entering the amounts to be paid not processed
budgetary expense adjustments	
(-)	Executed expense by entering the amounts to be paid not processed
(+)	Amounts to be paid not processed liquidated in the fiscal year
(-)	Previous fiscal years expenses
(-)	Formation of stock
(-)	Granting of advance money.
(-)	Not effective capital expenses
Equity adjustments	
(+)	Stock consumption
(+)	Expense incurred of advance money
(+)	Depreciation / amortization / depletion
= Cost	

The necessity of adjustment mostly reflects the current fragility of accounting, still very influenced by budget control instead of asset control.

Previous Fiscal Years Expenses

In the public sector, the legislation recognizes the possibility of situations where the expenses relative to previous fiscal years might be committed, liquidated and paid for in the current fiscal year.⁵ These expenses are recorded in clearly segregated accounts and must be expurgated, if one wants to know the cost of services of the fiscal year.

At last, it is necessary to understand that “costs of the period” exist which are not included in expenditure and or in “liquidated values”, classified as costing expenses, since these are not a part of the budgetary system, such as depreciation of fixed assets and labor charges, such as paid licenses. It is necessary that this information is registered in the patrimonial system of governmental accounting and incorporated to the costs of services rendered, as the current norms of accounting applied to the public system determine.

4. Relationship between public policies, resources, activities and cost objects

The direct relationship between public policies, resources, activities and cost objects can be seen in figure 3. This figure shows in short that the definition of public policies is the main determinant of public sector costs, deriving from that the programs and actions, as well as the identification of the necessary resources to execute them. The estimated and consumed resources are accumulated directly on projects and activities. Starting from accumulated costs from projects or activities, by a process of aggregation, it is possible to find the costs of functions and responsibility centers. The unit costs of products or services derive from the relationship between the cost of activities and the amount of physical quantities executed.

Some questions might be posed to this proposal. The first is that the costs gathered in this manner might contain distortions, “the most significant one being the lack of judicious

⁵ See art. 37 of law n° 4.320/64

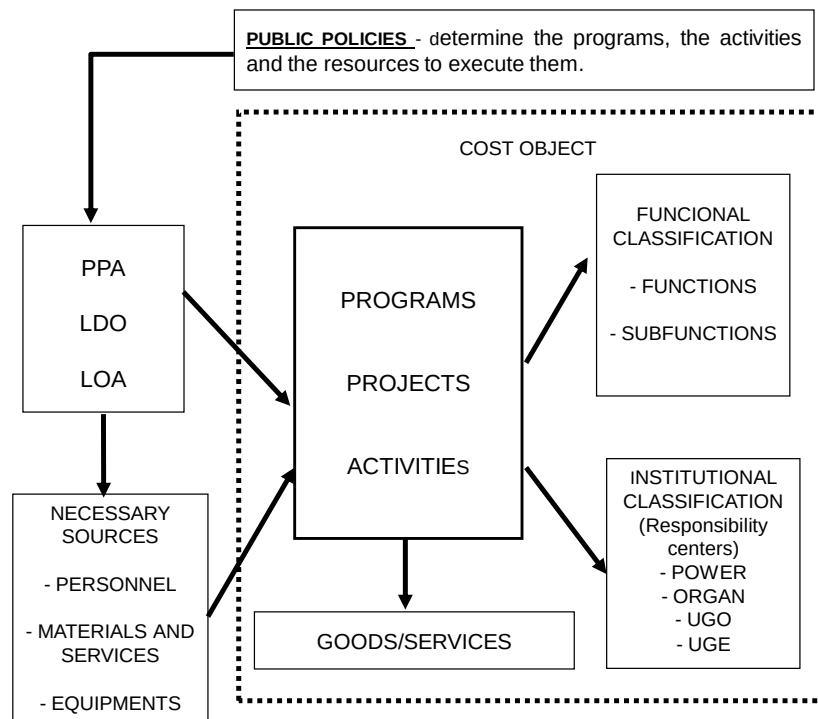
criteria in the linking of personnel to projects and activities they actually operate”. However, it is important to highlight that the quality of the integrated system’s information fundamentally depends on the accuracy with which resources are associated with activities and projects in the budget phase of the process and, afterwards, in the execution phase, depends on the rigor of the accounting entry using the correct classifications for each transaction.

Figure 3

Relationship: public policies, resources, activities and cost objectives

SYSTEM OF PUBLIC SECTOR COSTS

RELATIONSHIP: public policies, resources, activities and cost objects



Moreover, the best assurance of the improvement of the cost system are the calculus and the disclosure of unit costs of goods and services given to the population, for these will force managers involved to analyze the information, hoping to, on one hand, correct the classification distortions and, on the other, make more efficient the use of resources. It is worth highlighting that determining costs must be a process of continuing learning and improvement, the information is being generated and when used and analyzed, it serves as feedback to the whole process. The maxim of “riding a bike” is also valid, it is best to have a

“little” theory and practice it and in that process keep improving the model and the information, than a model which is ideal only in paper. This philosophy was fundamental so that in March 2010 the CIS was approved and later available for use.

The second question is that activities and projects are excessively generic or global, not allowing a direct association of consumed resources with its products. Here it is necessary to understand the diversity of governmental actions. There are cases in which the activity, even if generic, has only one final product, delivered to the citizens or users of public services. In that case, the cost appropriation to the product occurs directly and uneventfully. When there is more than one product resulting from the same activity, it can be treated as joint production (that is one of the techniques). That is, indicating the total cost of the activity and the amounts produced of each different product. The efficiency of the manager would be analyzed comparing the effectively produced with the pattern or the budgeted.

On the other hand, one cannot forget that the list of projects and activities is a prerogative of each one of the public entities; so, it is possible to create a dictionary of more generic activities or more specific according to the needs of information. That notwithstanding, surely there will be activities whose products will be of difficult explicitness, enumeration and quantification; in that case, obviously, there will be some difficulties in the calculus of units cost, whichever costing method is adopted.

This multiplicity and complexity were conceptually and methodologically addressed adequately, resulting in the modeling of CIS in two levels. One should also point out some of the main conceptual characteristics applied to CIS.

5. Accumulation System and Costing Method in the Public Sector

In this section the concepts of accumulation systems and costing methods are segregated, as well as its applicability in the public sector.

Accumulation System of the Public Sector

Typical services of the public sector, such as education, medical care, justice distribution, security and national defense, and others, are provided in a continuum, requiring, therefore, a system that accumulates costs by processes. That means that the costs of these services must

be accumulated during a certain period – month, trimester, semester -, the unit cost being calculated by the comparison between the cost of the period and the units produced in the same period. In the end of the accounting period, all accounts that accumulate costs must be closed.

On the other hand, the public sector also creates capacity to perform services when it builds a school, a hospital, a bridge, housing, or a road. That kind of action characterizes itself as specific order, with initial and ending dates for execution, meaning, therefore, that the cost accumulation system needed in this case is the “service order” one. The costs relative to this work must be accumulated in this order until the work is over, when total cost or the units costs of the order will then be calculated, when that is the case. One should highlight that the end of the fiscal year does not end the order; the costs relative to the ongoing order must be activated as ongoing work or any similar title.

Having in mind the present conceptualization for projects and activities in the public sector, one can conclude that the cost of the project must be accumulated by order, and the cost of activities, accumulated by process.

Costing System of the Public Sector

The costing system of the public sector rests on the historic and estimated costs. The historical cost is a fundamental element in the integration between the cost system and the public accounting and budgeting systems. In turn, the estimated cost may rest in simple estimates or use the standard cost tool. The ideal is that patterns for all activities and products to be executed by the public sector were created; however, that is not feasible in the beginning of the process of implementing costs, but an ideal to be pursued. It is important, however, that the cost information system creates conditions to specify, in the estimated cost, when it is about standard or mere estimative.

Costing Method in the Public Sector

The choice of costing method in the public sector is not an easy task, to the extent that there are advocates for almost all methods. Besides, in the public sector, unlike the private sector, there is no legal limitation regarding to the use of costing methods. In that sense, any one of the five methods (full costing, by absorption, costing activities, variable costing and direct) may be adopted.

Direct Costing Method in the public sector

The first argument, in favor of direct costing, is that this method, for not having any kind of apportionment, permits the analysis of the performance of managers and public policies with more objectivity and without the never-ending discussions about the general costs transferred. And facilitating the analysis of results and performance is the main goal of the Integrated Costs System.

The second argument is that this method is deep-seated in the systems of planning, budget and accounting of the budgetary execution, which facilitates the implementation of the integrated cost system.

This fact can be demonstrated in the following ways:

- Governmental action – projects and activities – is the category used for the programming of budgetary expenses;
- The resources necessary to the execution of these actions, classified according to their economical nature, are estimated and detailed by project and activities. So, the resources to be consumed associate themselves directly with the activities to be executed;
- The governmental action develops in a certain organizational space which is perfectly identifiable, being, therefore, possible to directly accumulate the executed expenses by activities/projects, by organs, by program managers or authorizers of expenses. As the projects/activities must serve a governmental function, they are also easily identifiable with the functional classification;
- The direct accumulation of budgetary expenses model may be transformed into a cost accumulation model of the period or order, once the adjustments of the liquidated

budgetary expenses and investments are done (mainly stock consumption), as it was discussed before in accounting conceptual adjustments.

The third argument is that the cost-benefit relation of information, however difficult it is to be measured, is clearly favorable to the direct costing method, since its implementation can be done rather quickly and with low training costs, since for employees and managers this method will ring a bell.

It is worth highlighting that the CIS proposal is to adopt as costing method the direct costing until the intermediate platform (“bus station”), conceptualized here as the method that allocates all costs⁶ -fixed and variable- directly to all cost objects without any kind of apportionment or appropriation. And from the “bus station” downwards (figure 1), the diversity and physical operational reality of each unit must be respected.

The choice of direct costing until the “bus station” (figure 1) guarantees the comparability and is corroborated in the literature, since the larger the complexity on the allocation schemes, the larger the induced classification and measurement mistakes. Thus, the proposed general model guarantees the possibility of comparison, without losing sight of customization. That is the biggest virtue of the model: living with both needs.

Costing by activities in the public sector

Another possible question is that the direct costing method proposed does not incorporate the advances in activity-based management. This management tool deals, fundamentally, with the unused capacity and activities that don’t aggregate value, seeking to reduce waste and eliminate unnecessary activities. At first sight, there would be no major difficulties in adopting activity based costing in the public sector, since:

⁶ Highlighted that governmental accounting, usually, does not register the depreciation of property and the amortization of financial expenses relative to long term debits, as discussed in Machado (2002).

- Every governmental action is decomposed in activities/ projects. The activities/ projects are a key element in the budget and accounting systems;
- The concept of public sector activity is very similar to the one formulated by Brimson:

Legal concept – an instrument of programming to achieve the objective of a program, involving a set of operations that take place in a continuum, from which a product necessary to the maintenance of governmental action results.

(Art. 2nd, law n° 42, April 14th 1999)

Brimson's concept (1996:63-64) – activities are processes that consume substantial resources to generate a production or even a basic criterion of activities based accounting is the capacity to relate an activity to a product, process, project or other objective the report which the administration needs cost information.

However, it is necessary to consider that the adoption of the activities-based costing method would amplify the difficulties to implement an integrated general system – cost, information and accounting – for many reasons:

- The process of implementation would require the identification of relevant activities, average time of execution of each one of them and number of hours each employee worked, amongst other aspects;
- The training would have to be intensive to standardize the criteria to be adopted in all organs or costing centers;
- The period of time necessary for the system to stabilize and produced benefits would be greatly broadened;
- The system would generate an amount on information practically impossible to be analyzed by the central managing organs (governmental offices, planning and finance ministries).

However, if the proposed cost system (direct costing) is implemented, nothing prevents –if it is even desirable – that, without any additional work, specific organs, from the information extracted from the accounting-management information system (up to the “bus station”, figure 1), build specific models (after the “bus station”, from it

downwards) of analysis and management of its own costs, following whatever other costing and appropriation model.

This gradual adoption (and, consequently, allocation), separated in two big moments – macro and micro –, constitutes an essential requisite for the cost model integrated with accounting, allowing the comparability to be done based on cost objects (figure 3) already used in the Brazilian culture of management.

Accordingly, working with the idea of hierarchy of activities, one can assume that the projects/activities correspond to the concept of macroactivities, which may be decomposed into smaller activities. From accumulated costs in activities/projects or in departments, drivers of resource consumption are identified by these smaller activities. Next, the drivers of the cost of these activities are identified with goods and services. During this process, the identification of unused capacity is also searched for, allowing the possibility of identifying the activities that don't aggregate value to goods or services.

6. Costs in the public sector and the government's information system

Summarizing the guidelines proposed in this article, figure 4 shows the relationship between the main actions of the ample decision model and the information system. The figure shows, also, the integration of the cost system of the public sector and the other information systems: budget and accounting.

It is interesting to observe that the cost information system absorbs the data of the accounting and budgetary systems. For that, it is necessary only to make the accounting adjustments, discussed previously, transforming expense into cost. Besides the data absorbed by these systems, the cost information system needs to register, only, the data relative to the goods and activities executed.

At last, it is emphasized the fact that cost objects, as it was mentioned, are aggregated using the same accumulation rules as in budgetary expenses.

Figure 4

Cost Information System of the Brazilian federal government and the other information systems

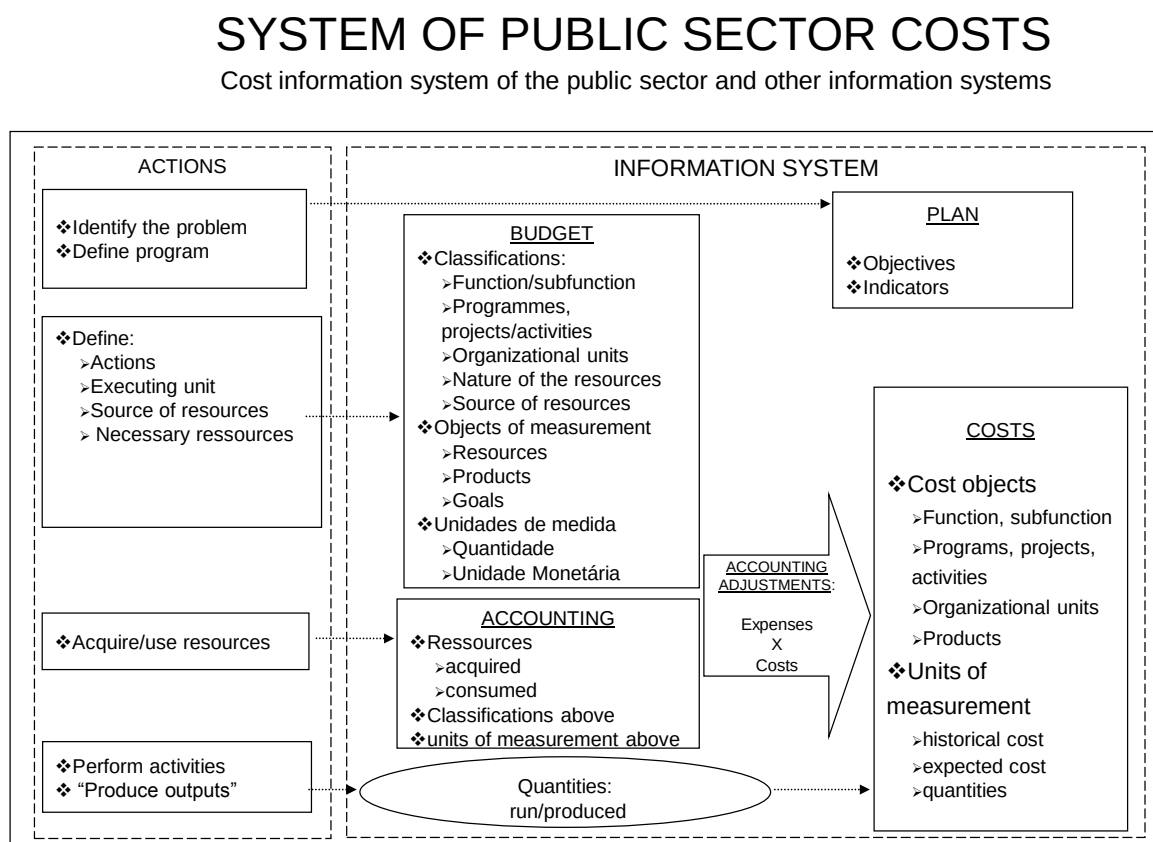


Figure 4 will not be described in detail, for the objective is to show the systemic vision that was prioritized in the conception model and implementation of CIS. The figure reinforces in other respects some details already evidenced in figure 1, mainly the integration of the Cost system to accounting and budget, from the conceptual adjustments necessary. A more analytical description of the whole model of integration, covering the whole cycle, will be the object of another article.

7. Final Considerations and Conclusion

The Cost Information System of the Brazilian federal government allows the effective measurement of costs from the administrative and programmatic viewpoint, from central organs of planning, budget, accounting and finance, to meet evenly the needs of all organs and entities in the federal structure. The methodology of cost measurement used coexists with the current limitations in public accounting, the lack of integration of the structuring systems and the lack of standard of organizational structures of the federal government. With the adoption of accrual accounting applied to the public sphere, this methodology will “gradually perfect itself”.

Reinforcing and highlighting the main guidelines

Some general principles must be considered to define the trajectory of implementation of a cost information system:

- The benefits brought by the information must surpass the cost necessary to obtain them;
- The information generated must be useful to the decision making process, which means having predictive power, allowing to anticipate, correct or confirm expectations;
- The information must be timely and reliable, in other words, represent the facts with fidelity, neutrality and in a verifiable way;
- Information must be comparable in time or between organizations, services and common cost objects.

Considering the construction of the cost system and its linking to the current budgetary, financial and accounting model and further considering the specificities of the proposed model from the central organ’s viewpoint, these general requisites are highlighted:

- The Cost Information System integrates itself conceptually and systematically to the subsystems: public budget, governmental accounting and data processing;
- These subsystems compose the accounting management information system of the public entity and has as a limit of performance the internal data and information;
- The public budget system is the repository for public policy decisions and the guide for its execution;

- The information about values and quantities budgeted, classified under institutional, functional and programmatic, are the starting point of the Cost Information System and, consequently, of the evaluation of managers regarding efficiency, efficacy and effectiveness;
- The budgetary liquidated expenses are equivalent to the concept of expenditure and can be considered cost of goods or services rendered in the period after the reclassifications and necessary conceptual adjustments;
- Program – main cost object of the system. That guideline stops the total standardization of the cost system's accounts;
- The set of cost objects is formed by categories of the budgetary system – programmatic, functional, institutional, and goods and services rendered, identified and measured in the budgetary system;
- Cost accumulation system – projects by order and activities by process. From the cost of the project and activities, the accumulation is done according to the hierarchy of accounts of the budgetary-accounting system;
- The costing system is based in historical and expected costs;
- Direct costing method – recommended until the “bus station” and from it forward according to the physical operational reality and the maturity of the management and the information systems necessary for the capturing of data to be used.

Using the cost information requires expansion of the management maturity

The cost information provides fundamental content to subsidize governmental decisions on more efficient allocation, facilitating sustainability and economic development. Its instrumentalization requires the transparency of processes and the definition of steps for the reduction of fragilities derived from management cycle and, consequently, of public policies. It is necessary to foment the debate over cost measurement and highlight its importance for the improvement of expense quality in the public sector and qualify the discussion about perspectives and challenges in cost information management in the public administration.

In a modern management model in managerial bases, one cannot (or should not) make decisions without knowing the different alternatives of action, its costs and benefits. Starting from the effectuation of the Cost Information System of the federal government the condition

for change was effectively potentiated. The simple existence of the system shows that an important step was taken, now it is necessary to foment debate on costs measurement and highlight its importance for the improvement of expense quality in the public sector, qualifying the discussion over the perspectives and challenges in cost information management in the public administration.

Future Challenges

It is important to highlight that to govern is not like moving the controls of a machine and obtaining the expected results. Besides from the definition of governmental areas, and its functions, questions arise at the same time regarding the way to act. Questions such as: with which resources, do directly or by delegation, with what priority, are examples. One must not lose sight that, in the Brazilian case, the challenge is double: the search for economical competitive edge fomented by an effective state and the reduction of social inequities. The state must prioritize the universalization of public policies. That is not, as we see it, contradictory with the efficiency of governmental action. As an example, one can mention the performance of public banks, which, even though they fulfilled very relevant functions in the resumption of credit during the world economic crisis, presented revenue and results far superior to those of the private sector.

In the future, from the advances in public accounting, through the adoption of accrual basis, the implementation of the cost system will allow, in the medium term, the increase of possibilities of improvement of the efficiency of the Brazilian governmental spending.

The discussions that deal with general norms (sometimes called “New Law of public finances”, sometimes called “Law on the quality of spending – LQF”) continue and have intensified in 2010. The referred to norm tries to discipline many guidelines for the elaboration of plan, budget, public accounting, debt levels and control. Many aspects relative to the responsibility in the budgetary process, in the accounting-financial and assets management, in order to strengthen the responsible management, are in course. It appears that the more mature the theoretical conception and the discussion around the conception is, the easier it will be to take this consensus to the legal framework. However, based on empirical disclosure, the

change will not be done by decree, continued improvement in the conceptual framework will be needed and, above all, changes in behavior.

Knowing about all the possibilities, the biggest challenge is to make the generated information useful and appropriate to the decision making process.

Bibliography

ALBUQUERQUE, Cláudio; MEDEIROS Márcio; FEIJÓ, Paulo Henrique. Gestão de finanças públicas. FEIJÓ, Paulo Henrique (Ed.). 2. ed. Brasília, 2008.

ALONSO, Marcos. Custo no serviço público. Texto para Discussão, n. 31. Brasília: Enap, 1998.

BRASIL. Lei nº 4.320/64, de 17 de março de 1964. Estatui normas gerais de direito financeiro para elaboração e controle dos orçamentos e balanços da União, estados, municípios e do Distrito Federal. Diário Oficial da República Federativa do Brasil, Brasília, March 23rd 1964.

_____. Decreto-lei nº 200, de 25 de fevereiro de 1967. Dispõe sobre a organização da administração federal, estabelece diretrizes para a Reforma Administrativa e dá outras providências. Diário Oficial da República Federativa do Brasil, Feb. 27 1967, rectified in Mar. 8th, mar. 30th and Jul 17th 1967.

_____. Lei Complementar nº 101, de 4 de maio de 2000. Estabelece normas de finanças públicas voltadas para a responsabilidade na gestão fiscal e dá outras providências. Diário Oficial da República Federativa do Brasil, Brasília, May 5th 2000.

_____. Lei nº 10.180, de 6 de fevereiro de 2001. Organiza e disciplina os sistemas de planejamento e de orçamento federal, de administração financeira federal, de contabilidade federal e de controle interno do Poder Executivo Federal, e dá outras providências. Diário Oficial da República Federativa do Brasil, Brasília, Feb. 7th de 2001.

BURBIDGE, John L. The principles of production control. London: Macdonald & Evans Ltd., 1971.

CRUZ, Flávio da et al. Comentários à Lei no 4.320. 3. ed. São Paulo: Atlas, 2003.

GARRISON, Ray H.; NORREN, Eric W. Contabilidade gerencial. Tradução de José Luiz Pavarato. 9. ed. Rio de Janeiro: LTC, 2001.

GIACOMONI, James. Orçamento público. 11. ed. São Paulo: Atlas, 2002.

GONÇALVES, M. A.; ZAC, J. I.; AMORIM, C. A. Gestão estratégica hospitalar: aplicação de custos na saúde. Revista de Administração FACES Journal, v. 8, n. 4, p. 161-179, 2009.

HILL, N. T. Adoption of costing systems in U.S. hospitals: an event history analysis 1980-1990. *Journal of Accounting and Public Policy*, v. 19, n. 1, p. 41-71, 2001.

HOLANDA, Victor Branco. Controladoria governamental no contexto do governo eletrônico — uma modelagem utilizando o enfoque sistêmico e a pesquisa-ação na Coordenadoria de Controle Interno da Secretaria da Fazenda do Estado de São Paulo, 2002.

_____; LATTMAN-WELTMAN, Fernando; GUIMARÃES, Fabrícia (Orgs.). Sistema de informação de custos na administração pública federal: uma política de Estado. Rio de Janeiro: FGV, 2010.

HORNGREEN, Chalers T. et al. Contabilidade de custos. 9. ed. Rio de Janeiro: LTC, 2000.

IFAC (International Federation of Accountants). Public Sector Committee. Perspectives on cost accounting for government. Study 12. New York, 2000. Available in: <www.ifac.org>. Access on: mar. 13th 2010.

KOHAMA, Heilio. Contabilidade pública: teoria e prática. 9. ed. São Paulo: Atlas, 2003.

KRAVCHUK, R. S.; SCHACK, R. W. Designing effective performance-measurement systems under the government performance and results act of 1993. *Public Administration Review*, v. 56, n. 4, p. 348-358, 1996.

LAGIOIA, U. C. T et al. A gestão por processos gera melhoria de qualidade e redução de custos: o caso da unidade de ortopedia e traumatologia do Hospital das Clínicas da Universidade Federal de Pernambuco. *Revista de Contabilidade e Finanças*, v. 19, n. 48, p. 77-90, 2008.

LEONE, George Sebastião Guerra. Custos, planejamento, implantação e controle. 3. ed. São Paulo: Atlas, 2000.

LUQUE, C. A. et al. O processo orçamentário e a apuração de custos de produtos e serviços no setor público do Brasil. *Revista do Serviço Público*, v. 59, n. 3, p. 309-331, 2008.

MACHADO, Nelson. Sistema de Informação de Custo: diretrizes para integração ao orçamento público e à contabilidade governamental. Brasília: Enap, 2005.

MACHADO JR., José Teixeira; REIS, Heraldo da Costa. A Lei 4.320 comentada. 30. ed. Rio de Janeiro: Ibam, 2000/2001.

MARTINS, Eliseu. Contabilidade de custos. 9. ed. São Paulo: Atlas, 2003.

MAUSS, César Volnei; SOUZA, Marcos Antonio de. Gestão de custos aplicada ao setor público. Modelo para mensuração e análise da eficiência e eficácia governamental. São Paulo: Atlas, 2008.

MIRANDA, G. J. et al. Custeio ABC no ambiente hospitalar: um estudo nos hospitais universitários e de ensino brasileiros. *Revista de Contabilidade e Finanças*, v. 18, n. 44, p. 33-43, 2007.

MOURA, José Flávio de Melo. O sistema de contabilidade do governo federal na mensuração dos custos dos programas de governo e das unidades gestoras. 2003. Dissertação (Mestrado) — Universidade de Brasília.

SHANK, John K.; GOVINDARAJAN, Vijay. Strategic cost management — the new tool for competitive advantage. New York: The Free Press, 1993.

SILVA, César Augusto Tibúrcio (Org.). Custos no serviço público. Brasília, 2004.

SILVA, Lino Martins da. Contribuição ao estudo para implantação de sistema de custos na administração pública. Brasília: Esaf, 1997.

_____. Contabilidade governamental — um enfoque administrativo. 5. ed. São Paulo: Atlas, 2002.